Week 2: Predicates

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- Predicates and quantifiers
- Universal and existence quantifications
- Properties of quantifiers
- Nested quantifiers and their orders

1.4 Predicates and Quantifiers

Definition

A **propositional function**, or **predicate**, is a proposition-valued function

$$P: \text{ a domain } \mathcal{D} \to \{\text{ propositions }\}$$

Example. $A(x)$: (Computer x is under attack by a hacker.)

A(1) : (Computer 1 is under attack by a hacker.)

...

A(17) : (Computer 17 is under attack by a hacker.)

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Example. $A(x)$: (Computer x is under attack by a hacker.)

Example. $P(x) : (x > 3)$ for x in $\mathcal{D} = \{\text{real numbers}\}$.
What are the truth values of $P(3.001)$ and $P(2.999)$?

(3.001 > 3) is T,
so P(3.001) = T

(2.999 > 3) is F,
so P(2.999) = F.

1.4 Predicates and Quantifiers

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Example. $A(x)$: (Computer x is under attack by a hacker.)

Example. $P(x) : (x > 3)$ for x in $\mathcal{D} = \{\text{real numbers}\}$.
What are the truth values of $P(3.001)$ and $P(2.999)$?

Example. $Q(x, y) : (x = y + 3)$ for x, y in $\mathcal{D} = \{\text{integers}\}$.
What are the truth values of $Q(1, 2)$ and $Q(7834, 7831)$?

Q(1,2) = (1 = 2 + 3) = F
Q(7834, 7831) = (7834 = 7831 + 3) = T.

Definition

The **universal quantification** of P(x) is the statement

"P(x) for all values of x (in a domain)",

or just "for all x, P(x)", and denoted by $\forall x, P(x)$.

All

Example.

- P(x) = (I come to the University on Day x) on $\mathcal{D} = \{$Mon, Tue, Wed, Thur, Fri$\}$.

  P(Thur) = (I come to the university on Thursday)

  "forall x, P(x)" means: "I come to the university everyday."

In calculus, (a,b) is an open interval, containing real numbers x where a < x < b.
[a,b] is an open interval, containing real numbers x where a ≤ x ≤ b.
Similar definitions apply to (a,b] and [a,b).

Example.

- "For all x in $\mathcal{D} = (3, \infty)$, $P(x) : (x > 3)$" is T.

“ For all x in D = (3, ∞), P(x): (x > 1)” is T.

Example.

- "For all x in $\mathcal{D} = (3, \infty)$, $P(x) : (x > 3)$" is T.

- "For all x in $\mathcal{D} = [3, \infty)$, $P(x) : (x > 3)$" is F.

  Since 3 is in D, P(3): (3 > 3) is F, even though P(x) is T for all other x in D.

Example.

- "For all x in $\mathcal{D} = (3, \infty)$, $P(x) : (x > 3)$" is T.

- "For all x in $\mathcal{D} = [3, \infty)$, $P(x) : (x > 3)$" is F.

Example.

{ positive integers } = {1,2,3,4,5, ….}

- "$\forall n$ in $\mathcal{D} = \mathbb{N}$, $Q(n) : (3n < 4n)$" is T

Example.

- "For all x in $\mathcal{D} = (3, \infty)$, $P(x) : (x > 3)$" is T.

- "For all x in $\mathcal{D} = [3, \infty)$, $P(x) : (x > 3)$" is F.

Example.

- "$\forall n$ in $\mathcal{D} = \mathbb{N}$, $Q(n) : (3n < 4n)$" is

{ non-negative integers } = {0, 1, 2, 3, 4, …}

- "$\forall n$ in $\mathcal{D} = \mathbb{N}_0$, $Q(n) : (3n < 4n)$" is F

$Q(0) = (0 < 0) = F$

Example.

- "For all x in $\mathcal{D} = (3, \infty)$, $P(x) : (x > 3)$" is T.
- "For all x in $\mathcal{D} = [3, \infty)$, $P(x) : (x > 3)$" is F.

Example.

- "$\forall n$ in $\mathcal{D} = \mathbb{N}$, $Q(n) : (3n < 4n)$" is
- "$\forall n$ in $\mathcal{D} = \mathbb{N}_0$, $Q(n) : (3n < 4n)$" is

  { all integers } = { … , −2, −1, 0, 1, 2, 3, 4, …}
- "$\forall n$ in $\mathcal{D} = \mathbb{Z}$, $Q(n) : (3n < 4n)$" is F

  Q(n) = F for all non-positive integers.

Definition

Given "$\forall x, P(x)$", If there is at least one c in $\mathcal{D}$ such that $P(c)$ is not true, then we say that c is a **counter-example** to "$\forall x, P(x)$".

Example.

- $x = 3$ is a counterexample to

$$\text{"for all x in } \mathcal{D} = [3, \infty),\ P(x) : x > 3 \text{ "}.$$

In this example, " for all x …., P(x)" is F

- There is no counterexample to

$$\text{"for all x in } \mathcal{D} = (5, \infty),\ P(x) : x > 3\text{"}.$$

In this example, " for all x …., P(x)" is T

Definition

The **existential quantification** of P(x) is the proposition

"There exists (at least) one x (in the domain) such that P(x)",

or just "there exists x, P(x)", and denoted by $\overset{\text{Exists}}{\exists} x, P(x)$.

Definition

The **existential quantification** of P(x) is the proposition

"There exists (at least) one x (in the domain) such that P(x)",

or just "there exists x, P(x)", and denoted by $\exists x, P(x)$.

Example.

- x in $\mathcal{D} = \{\text{teachers}\}$, $P(x) = (x$ teaches Discrete Math).
  in XMUM
  "there exists x, P(x)" means:

  We can find one teacher who teaches Discrete Math.

  "there exists x, P(x)" is T, because P(Geo Tam) = T
  even though P(any prof in Chinese Medicine) = F

Definition

The **existential quantification** of P(x) is the proposition

"There exists (at least) one x (in the domain) such that P(x)",

or just "there exists x, P(x)", and denoted by $\exists x, P(x)$.

Example.

- x in $\mathcal{D} = \{\text{teachers}\}$, $P(x) = (x$ teaches Discrete Math$)$.

  "there exists x, P(x)" means:

  "there <u>exists</u> a <u>unique</u> x, P(x)" means:

  there is one only one

  We can find one <u>and only one</u> teacher who teaches Discrete Math.

  exactly one

  "there exists a unique x, P(x)" is F, because P(Geo Tam) = P(Chris Chung) =T

Example.

- If x is in $(-\infty, 2]$, then"$\exists x,\ P(x) : (x \leq 2)$" is T.

- If x is in $(-\infty, 2]$, then"$\exists x,\ P(x) : (x = 4)$" is T.

  We can find 1 in D = (–∞, 2] , such that P(1) = T
  (even though P(x) is F for all other x in D)

- If x is in $(-\infty, 2]$, then"$\exists x,\ P(x) : (x > 4)$" is F.

  In fact, for all x in (–∞, 2], " P(x):(x > 4)" is F

Example.

- If x is in $(-\infty, 2]$, then "$\exists x,\ P(x) : (x \leq 2)$" is T.
- If x is in $(-\infty, 2]$, then "$\exists x,\ P(x) : (x = 1)$" is T.
- If x is in $(-\infty, 2]$, then "$\exists x,\ P(x) : (x > 4)$" is F.

Sometimes $\exists!$ is used if there is a unique x such that $P(x)$.

Example.

- If x is in $[2, \infty)$, then "$\exists! x,\ P(x) : (x \leq 2)$" is T

  We can find x = 2 in D = [2, ∞), such that P(2) is T

  AND this x is the only x in D making P(x) = T.
- If x is in $[2, \infty)$, then "$\exists! x,\ P(x) : (x > 4)$" is F.

  We can find more than one x, as long as x is in (4, ∞), such that P(x) = T.

  These x are not unique, so " ∃! x, P(x) : (x > 4) " is F.

Example. Put $Q(x) :\ (x = x + 1)$, then "$\exists x, Q(x)$" is F.

We cannot find such x that "x = x+1"

Quantifier over finite domain.

$$\mathcal{D} = \{1, 2, \ldots, n\}$$

Relation between $\wedge/\vee$ and $\forall/\exists$, when $\mathcal{D}$ is finite:

- "$\forall x$ in $\mathcal{D} = \{1, 2, \ldots, n\}, P(x)$" is equivalent to
$$P(1) \wedge P(2) \wedge \cdots \wedge P(n).$$
- "$\exists x$ in $\mathcal{D} = \{1, 2, \ldots, n\}, P(x)$" is equivalent to
$$P(1) \vee P(2) \vee \cdots \vee P(n).$$

Remark

If $\mathcal{D}$ has infinite numbers (such as an interval of real numbers), we cannot use $\wedge/\vee$. We can only use $\forall/\exists$.

Precedence of Logical Operators

0. $\forall$ and $\exists$ Priority
1. $\neg$
2. $\wedge$ and $\vee$
3. $\rightarrow$ and $\leftrightarrow$

Example. $\forall x, P(x) \vee Q(x)$ means $\underbrace{\text{(forall x, P(x))}}_{\text{proposition}}$ or $\underbrace{\text{Q(x)}}_{\text{predicate}}$

"$\forall$x, P(x) $\vee$Q(x) " is wrong.
We need to write: $\forall$x, (P(x) $\vee$ Q(x))

In the statement (P(x) $\vee$ Q(x)), both P(x) and Q(x) are viewed as propositions with an input x.
P: D —> {propositions}, then P is viewed as a predicate.

Some equivalences

$$\forall x\ (P(x) \land Q(x)) \equiv (\forall x\ P(x)) \land (\forall x\ Q(x),)$$
$$\exists x\ (P(x) \lor Q(x)) \equiv (\exists x\ P(x)) \lor (\exists x\ Q(x).)$$

(similar to the distributive law)

Some equivalences

$$\begin{aligned} \forall x\,(P(x) \wedge Q(x)) &\equiv \forall x\, P(x) \wedge \forall x\, Q(x), \\ \exists x\,(P(x) \vee Q(x)) &\equiv \exists x\, P(x) \vee \exists x\, Q(x). \end{aligned}$$

Think of the finite domain case: D = {1,…, n}, or even D = {1,2}

When D = {1,2}, $\forall$x in {1,2}, (P(x) $\wedge$Q(x)) = (P(1) $\wedge$Q(1)) $\wedge$ (P(2) $\wedge$Q(2))

= P(1) $\wedge$Q(1) $\wedge$ P(2) $\wedge$Q(2) (by associative law)

= P(1) $\wedge$ P(2) $\wedge$Q(1) $\wedge$ Q(2) (by commutative law)

= (P(1) $\wedge$ P(2)) $\wedge$(Q(1) $\wedge$ Q(2)) (by associative law again)

= $\forall$x P(x) $\wedge\forall$x Q(x),